IntrusionX Second Edition

Kavach AI

TEAM NAME: ASTRA

THEME: AIML

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KAVACH AI – Real-Time Document Trust & Fraud Prevention System

Kavach AI is a real-time AI system that verifies the authenticity of digital documents and detects manipulation before institutional approval.

AI Tampering Detection (CV + OCR)

Trust Score + Fraud Heatmap

Secure Document Fingerprinting

Scalable API Integration

HOW IT ADDRESSES THE PROBLEM

- Eliminates slow manual verification and reduces rising document fraud.
- Enables fast, accurate decisions through real-time and self-learning AI

INNOVATION & UNIQUENESS

- Template+ MetadataIntelligence for advanced fraud detection.
- Self-Learning AI improving from new fraud patterns.
- Batch Processing + REST API for scalable institutional use.
- Explainable AI with TrustScore insteadof binary detection.

TECHNICAL APPROACH

Step 1 – Document Input

Upload document (Image / Scan / PDF)

Step 2 – Preprocessing

- Image normalization
- Noise reduction
- Format standardization

Step 3 – AI Fraud Analysis

- Tampering detection (Computer Vision)
- OCR-based text extraction
- Template consistency verification
- Metadata anomaly detection

Step 4 – Risk Scoring Engine

- Multi-factor fraud scoring
- Trust Score generation

Step 5 – Output Visualization

Fraud heatmap display

Step 6 – Document Fingerprinting

Technologies to be Used

- ComputerVision (CNN / Vision Transformers) – Tampering detection
- OCR (PaddleOCR / Tesseract) – Text extraction
- NLP Models – Logical anomaly detection
- Siamese Networks – Template verification
- Metadata Forensics – Edit history detection
- Backend – FastAPI / Node.js
- Frontend – React.js
- Database – MongoDB / Firebase

FEASIBILITY AND VIABILITY

Feasibility of the Idea

- Uses existing proven AI technologies (CV, OCR, NLP)
- Can be developed as a working prototype within hackathon timeline
- Scalable architecture for fintech & HR integrations
- Supports real-world document formats (scan, camera, PDF)
- Deployable as cloud-based verification API

Potential Challenges & Risks

- Detecting highly sophisticated document manipulations
- Variations in document layouts across institutions
- Limited real-world fraud datasets for training
- Maintaining high accuracy in diverse input conditions

Strategies to Overcome Challenges

- Multi-layer AI verification pipeline for robust detection
- Template intelligence model trained on diverse document samples
- Synthetic fraud dataset generation for improved training
- Continuous learning fraud pattern engine
- Human-in-the-loop verification for critical cases

IMPACT AND BENEFITS

Potential Impact on Target Audience

- 🏦 Reduces financial fraud in loan & KYC processes
- ✓ Enables instant document verification for fintech & banks
- 👥 Prevents fake hiring through certificate validation
- 🛡️ Strengthens trust in digital onboarding systems
- 📈 Scalable solution for startups, enterprises & government

Benefits of the Solution

Economic Impact

Reduces fraud losses worth crores in financial systems

Saves cost of manual verification processes

Increases efficiency in loan approvals & hiring

Social Impact

Promotes fairness in hiring & financial access

Prevents misuse of fake identities

Builds trust in digital ecosystems

Technological Impact

Introduces AI-driven trust infrastructure

Enables real-time fraud detection systems

Scalable across industries (fintech, HR, insurance)

RESEARCH AND REFERENCES

- Details / Links of the reference and research work

- Research Portal (Project Data & Insights):
- https://kavachaiinfo.netlify.app
- Global Identity Fraud Trends – Sumsub Fraud Report 2024
- Digital Document Forgery Growth – Global Banking & Finance
- Synthetic Identity Fraud Impact – McKinsey Research
- OCR & Computer Vision – IEEE Research Library
- Document Tampering Detection – Springer AI Research
- AI Verification Systems – Deloitte Insights